Assignment 01 (Individual) – GAMES DEVELOPMENT

SE4031 – Games Development

ABYSSAL DRIFT

Game Documentation

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Module	SE4031 - Games Development
Submission	Assignment 01 (Individual) - 2D Interactive Game

Win Condition: Reach a target score of 100 while protecting the submarine's hull integrity.

1. Game Title

Abyssal Drift

2. Summary of the Game Concept

Abyssal Drift is a 2D underwater side-scrolling survival game in which the player controls a small submarine travelling through deep-sea coral trenches. The player must steer the submarine vertically through narrow tunnel gaps, avoid coral collisions, defeat jellyfish enemies with sonic pulse attacks, and collect oxygen tanks to recover health. As the dive continues, the submarine moves faster, which increases tension and difficulty.

The game uses a clear objective as well as a survival challenge. The player wins by reaching a target score of 100 before the submarine's hull integrity reaches zero. If the hull integrity drops to zero first, the game ends and a Game Over screen is displayed.

3. Description of How Core Features Are Implemented

Health System

The player uses a Hull Integrity system with a maximum health value of 100. The current health is shown on the UI as a health bar. Colliding with coral obstacles reduces health, and colliding with jellyfish enemies causes higher damage. The player only stops when health becomes zero. On restart, health resets back to the maximum value.

Score System

The score is shown on the UI and increases when the submarine successfully passes through the coral tunnel gap. Additional score is awarded when jellyfish enemies are destroyed by sonic pulse projectiles. The score resets to 0 when the game restarts.

Health Packs

Health packs are represented as oxygen tanks. They are collidable pickup objects that restore the player's health up to the maximum value. When the submarine touches an oxygen tank, the object is destroyed. If the player passes it without collecting it, it disappears automatically after a short duration.

Projectile Attacks

The submarine fires sonic pulse projectiles using the left mouse button. Projectiles spawn from a FirePoint attached to the front of the submarine, move forward, and are destroyed after a short time or on collision. When a projectile hits a jellyfish enemy, the enemy is destroyed and score is added.

Enemies

Enemies are represented as jellyfish. They are visually different from the oxygen tanks and behave as damaging obstacles. Touching an enemy damages the player, while hitting the enemy with a projectile destroys it. If the player passes an enemy, it is automatically removed after a short duration.

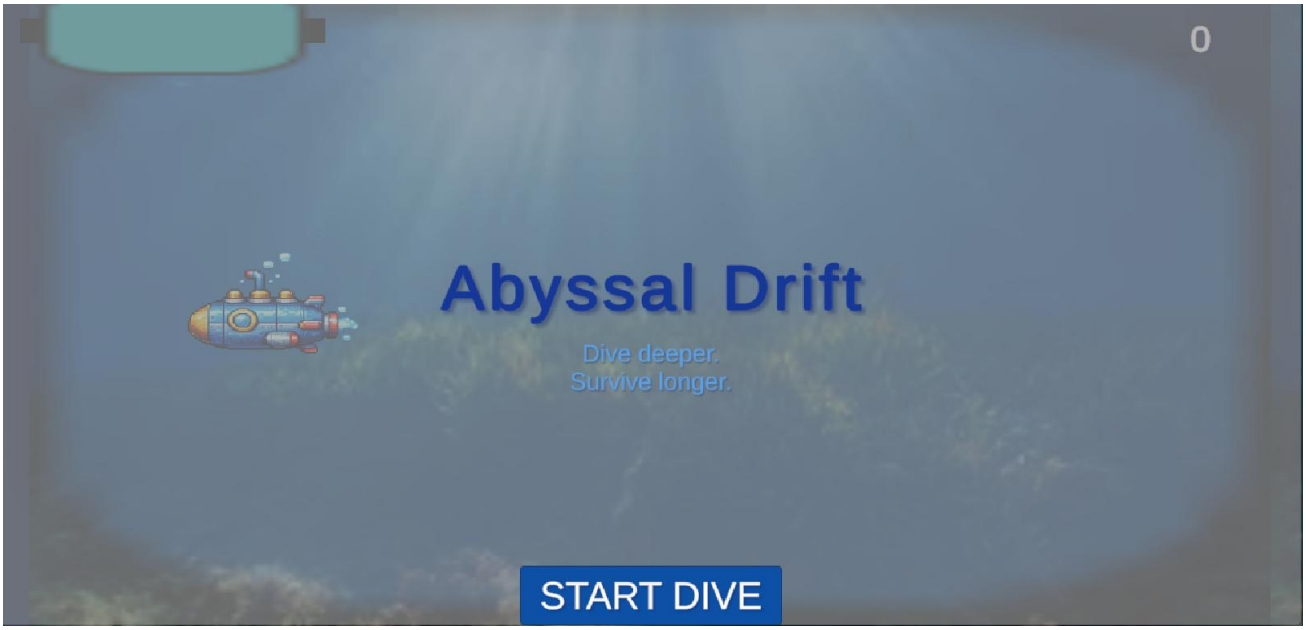
Increasing Speed

The forward movement speed increases gradually over time to make the game more challenging. A maximum speed limit is used so the difficulty remains fair and playable.

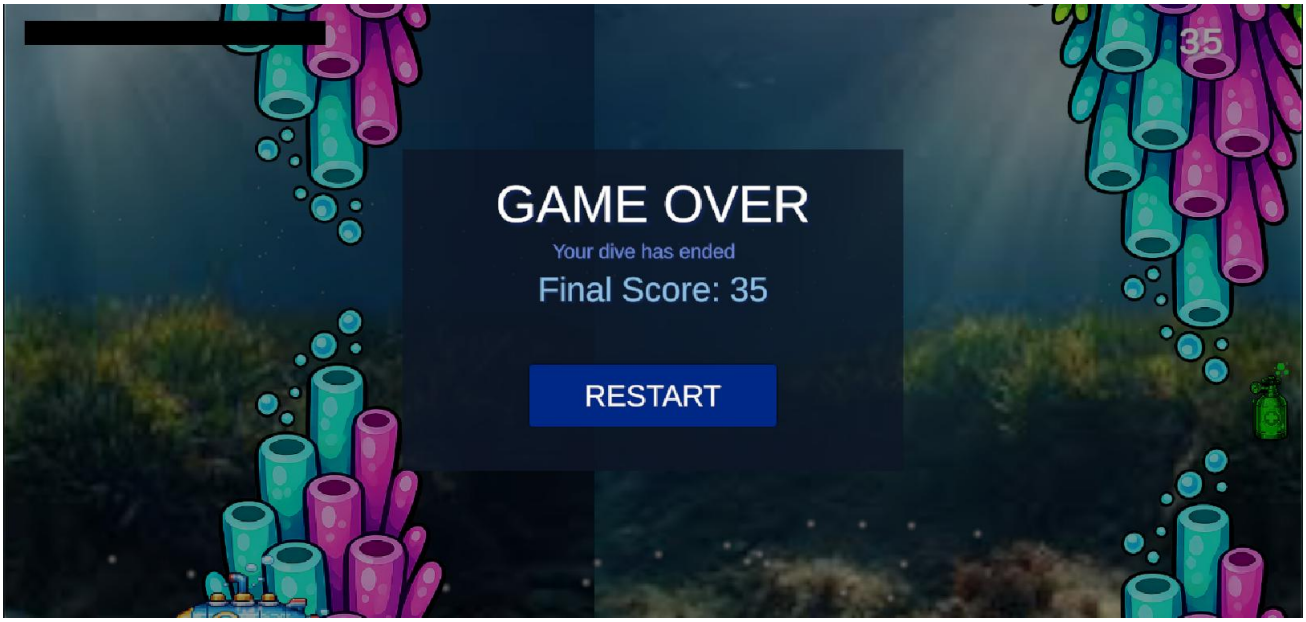
Menus and End States

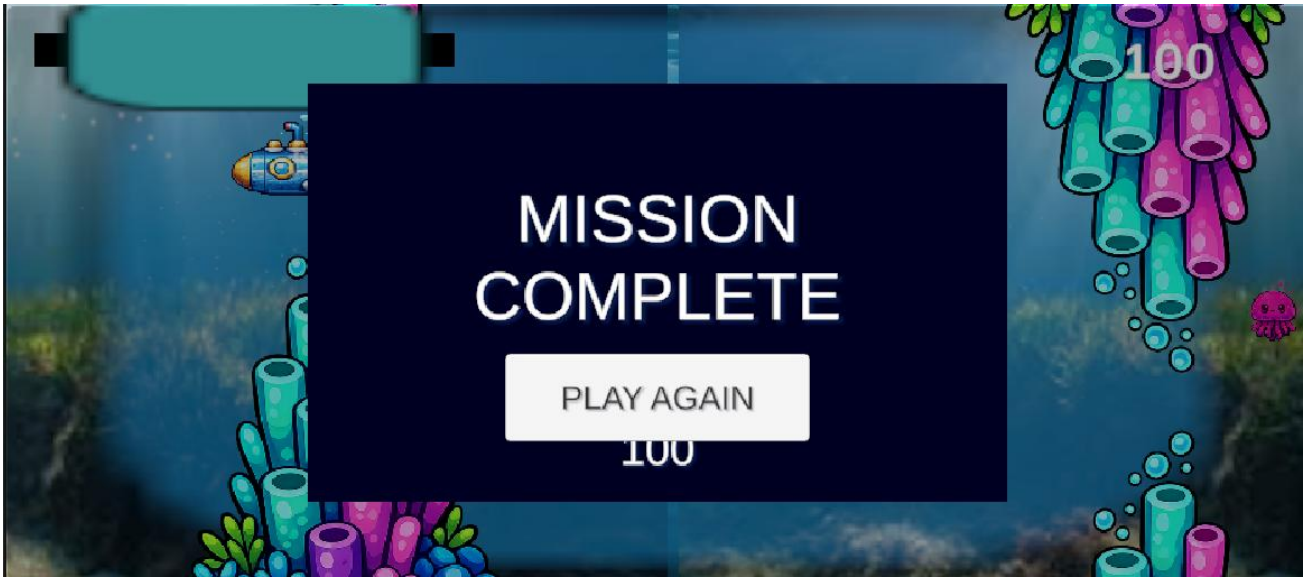
The game includes a Start Menu, Game Over screen, Restart Menu and Win screen. The Start Menu allows the player to begin the dive, the Game Over screen appears when health reaches zero, and the Win screen appears when the player reaches the target score of 100.

4. Screenshots of Gameplay



10
RESUME
PAUSE


35
GAME OVER
Your dive has ended
Final Score: 35
RESTART



5. Control Guide

Input	**Action**
W / Up Arrow	Move the submarine upward
S / Down Arrow	Move the submarine downward
Left Mouse Click	Fire a sonic pulse projectile
Play Button	Start the game from the title screen
Restart Button	Restart after Game Over or Win

6. Description of the Creative Feature

The main creative feature in Abyssal Drift is the dynamic underwater tint system. A transparent WaterTint overlay changes its color based on the player's remaining health. When health is high, the ocean appears brighter and clearer. As health decreases, the water gradually becomes darker and more ominous. This effect improves immersion while also giving the player an immediate visual indication of danger.

Additional creative polish includes parallax underwater backgrounds, bubble particle effects behind the submarine, floating plankton particles in the scene, and a sparkle effect when oxygen tanks are collected.

7. Credits for External Assets / Tools Used

Item	Details
Game Engine	Unity 2022
Programming / Editing	Visual Studio Code
UI Text	TextMeshPro
Sprites / Assets	Submarine, coral, jellyfish, and oxygen tank sprites generated and adapted for this project
Effects	Unity built-in particle system for bubbles, plankton, and pickup effects
Audio	Background underwater music

8. Gameplay Video Link

Video Link: Demo_video.mp4

Onedrive link of the game: SE4031_Assignment01_IT22641410

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